

Performance Testing

Date	12 July 2026
Team ID	SWTID-2026-6495
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. Complete the tests the applications response time, resource usage, and overall stability.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Swagger UI
Type of Testing	Functional & Performance Testing
Target Modules / API	Data Preprocessing, Model Prediction, Result Generation API
Test Environment	Local FastAPI + Streamlit
Test Duration	10 Minutes

Step 2: Test Scenarios

S.No	Test Scenario Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Upload applicant data for prediction	1	5	Data uploaded and processed successfully
2	Concurrent prediction requests	5	8	Predictions returned without failure
3	Model inference and result check	1	5	Model returns valid approval/rejection result

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Within limits
2	Response Time (Max)	< 3 seconds	2.8 sec	Pass	Slightly high under load
3	Throughput (Requests/sec)	> 10 req/sec	12 req/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 70%	55%	Pass	Normal usage
6	Memory Utilization	< 70%	60%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings:

- All core functions (data upload, prediction, result generation) responded within acceptable time limits.
- System handled concurrent requests without errors or crashes.

Bottlenecks Identified:

- Slight increase in response time observed during concurrent prediction requests.

Optimization Steps Taken:

- Optimized API request handling to reduce processing overhead.
- Improved model loading to reduce inference latency under load.

Step 5: Screenshots / Evidence

[Insert screenshots of your Postman/Swagger UI test runs, response times, and terminal logs here.]